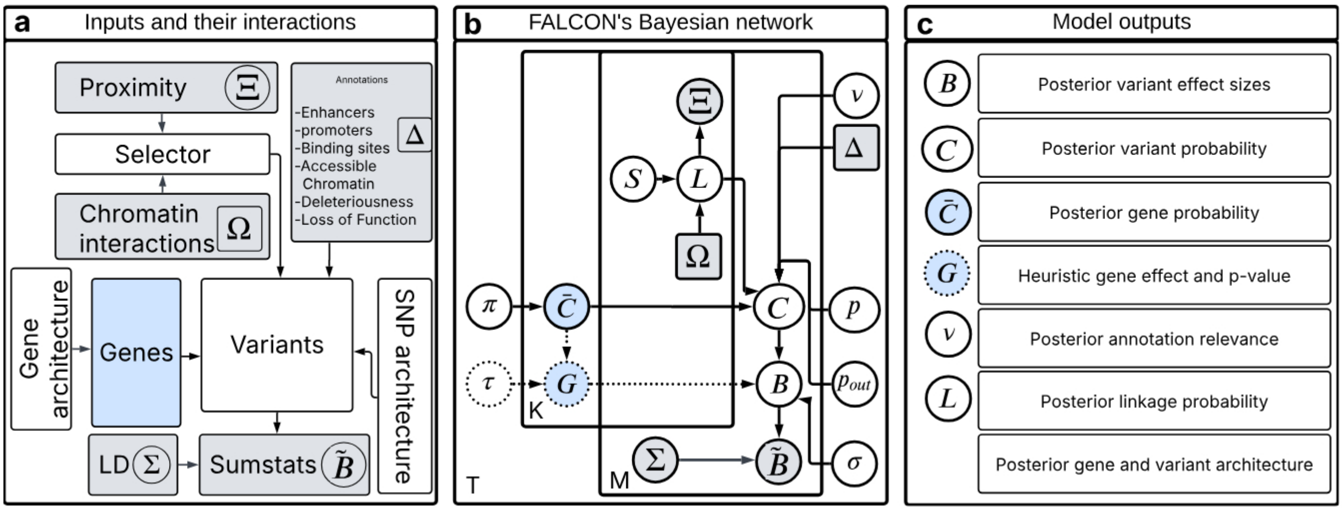
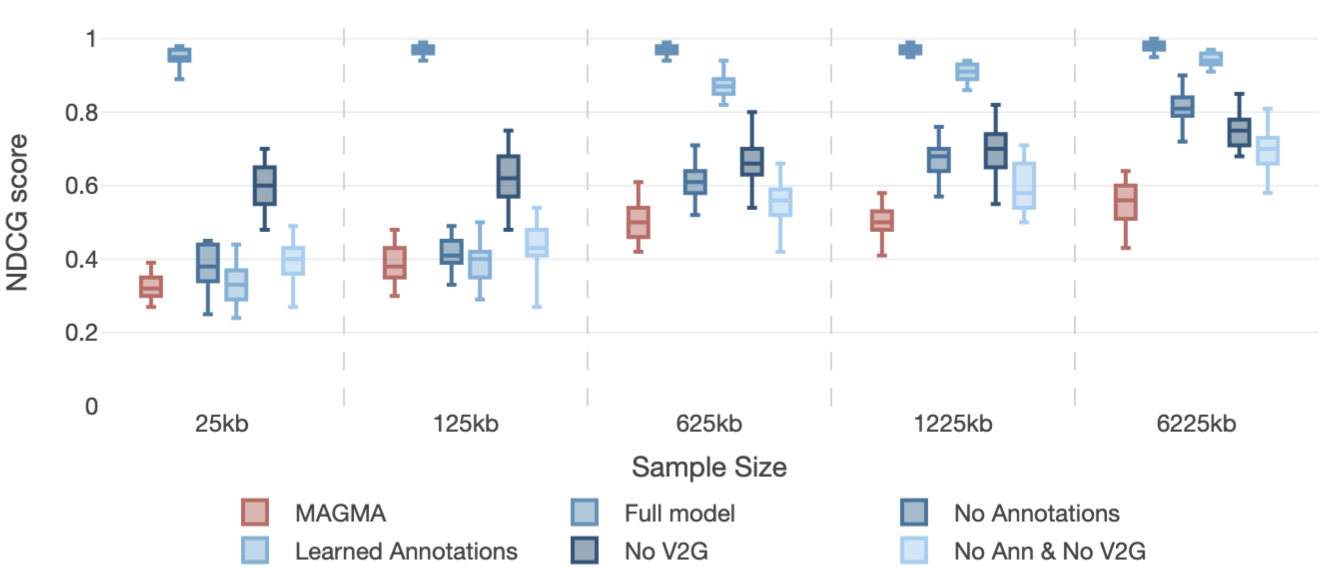
FALCON: Model, Simulations and Validation

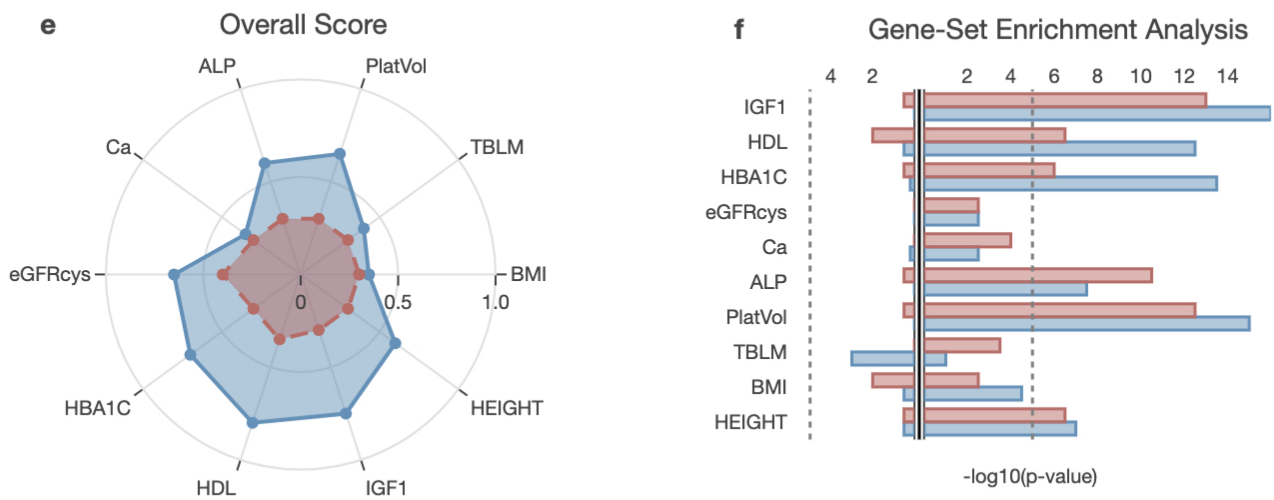
Overview of FALCON's model for the analysis of genetic support.



d Simulation and impact of genetic evidence.



Validation of gene prioritization and biological pathway identification on real data.



a, outlines the diverse data types integrated by the model. The primary inputs include summary statistics (can be GWAS or ExWAS), linkage disequilibrium patterns, functional annotations such as enhancer, promoters, binding sites and accessible chromatin locations for GWAS or deleteriousness and loss of function for ExWAS, and chromatin interaction scores that link regulatory regions to genes. These data sources provide the evidence used to infer association. **b,** FALCON's core is represented as a Bayesian network, a graphical model that illustrates the probabilistic dependencies between variables. Observed variables (shaded) represent the input data from panel (a). Latent variables (unshaded) represent the unobserved quantities the model aims to estimate. Inference within this network involves two types of information flow: Predictive propagation where information flows from a parent node to a child. For example, functional annotations inform on the probability that a variant is causal. And retrospective propagation a bottom-up flow where observed data at a child node updates beliefs about its parent. For example, the observed summary statistics are used to infer the variant's effect size. **c,** lists the primary outputs generated by FALCON, which are the posterior probabilities of the latent variables. These posteriors quantify the model's confidence in its inferences, such as the probability that a specific variant is the causal driver of an association signal (posterior variant probability) or that a particular gene is associated to a trait (posterior gene probability). **d** The figure compares the performance of FALCON (blue) and MAGMA (red) in identifying causal genes across simulated datasets. The performance is measured by the NDCG score (y-axis) at different sample sizes (x-axis, representing thousands of individuals). a, Full model: When FALCON is provided with the correct ("true") functional annotations, it dramatically outperforms MAGMA across all sample sizes. b, No Annotations: annotations are removed and V2G + Weibull is kept. In this scenario FALCON's performance decreases significantly. This highlights the critical role annotations play in improving accuracy. c, Learned Annotations: This panel shows FALCON's ability to learn annotation relevance from data. While not as powerful as using true annotations, this feature allows FALCON to partially recover its performance and still maintain a clear advantage over MAGMA, especially at larger